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Predictive Modeling of Petrophysical Properties in Lake-Facies Carbonates: A Case Study of Qaidam Basin

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Abstract

This study investigates the petrophysical attributes of the Nanyinshan reservoir situated in the Qaidam Basin, an area predominantly characterized by lake-facies carbonate rocks, emphasizing the predictive modeling of porosity using machine learning techniques. The geological analysis reveals the complex depositional systems and tectonic influences shaping the reservoir, with data from the Yi Ping 1 well serving as the foundation for this research. This area is predominantly composed of lake-facies carbonate rocks. A comparative evaluation of six machine learning algorithms, including Random Forest (RF), XGBoost (XGB), Support Vector Regression (SVR), Ridge Regression (RR), Polynomial Regression (PR), and Linear Regression (LR), were conducted to predict porosity. Among these, RF emerged as the most effective model, achieving the highest predictive accuracy ($R^2 = 0.9046$) and the lowest error metrics (Mean Squared Error(MSE) = 0.93, Root Mean Squared Error(RMSE) = 0.97, Mean Absolute Error(MAE) = 0.81). Residual error analysis highlighted RF's superior consistency, characterized by a narrow interquartile range and minimal bias, while simpler models like LR struggled with significant variability and sensitivity to outliers. Advanced models, such as Random Forest (RF) and XGBoost (XGB), demonstrated a robust ability to capture the non-linear relationships inherent in the dataset, outperforming traditional approaches. The results lay the groundwork for the integration of predictive analytics into reservoir management practices, mitigating uncertainties in resource assessment and enhancing exploration and production strategies. Subsequent research will aim to integrate further geological and geophysical datasets to enhance model accuracy and applicability.

Keywords: Qaidam basin, Petrophysical properties, Porosity, Machine learning, Tight carbonate

Introduction

Energy security remains a critical issue in the global quest for sustainable development. With increasing energy demand and geopolitical uncertainties affecting fossil fuel supplies, the efficient utilization of domestic resources has become an essential strategy for many countries. The advent of the North American shale-oil revolution has redirected the global oil and gas exploration focus towards unconventional resources. This includes mixed carbonate-siliciclastic fine-grained rocks, volcanic rocks, tight sandstones, and shales, which have become the new frontiers in hydrocarbon recovery (Macquaker et al., 2014; Zou et al., 2013). China, ranking third globally in uncoventional reserves, trails only Russia and the United States (Jia et al., 2016; Wang et al., 2021). The nation's abundant resources are dispersed across various basins, with notable examples including the Lucaogou Formation in the Junggar Basin, the Shahejie Formation in the Bohai Bay Basin, and the Qingshankou Formation in the Songliao Basin. The Qaidam Basin, distinguished as the only large-scale Cenozoic lacustrine mixed carbonate-siliclastic rocks, identified as oil-bearing basin in western China, hosts a unique saline lake basin oil-gas system (Song et al., 2010, 2014; Jian et al., 2014; Ji et al., 2017). Within the Western Qaidam Depression, several oil-bearing strata have been identified, ranging from the Upper Xiaganchaigou (~38—22 Ma) to Xiayoushashan Formations (~14.9—8.2 Ma) (Ma et al., 2018; Guo et al., 2020). PetroChina's Qinghai Oilfield Company's exploration efforts have established that these low-permeability rocks serve dual roles as hydrocarbon sources and reservoirs (He et al., 2008; Feng et al., 2013).

The Nanyishan oilfields, situated in the northwestern Qaidam Basin, have been significant in this context, with oil and gas primarily occurring in both deep Paleogene and shallow Neogene mixed carbonate-siliciclastic sequences (Zhang et al., 2006; Feng et al., 2011; Wang et al., 2012). Reservoir characterization indicates that deep reservoir formation is influenced by structural fracturing and diagenetic dissolution, while shallow reservoir spaces are characterized by primary pores in the mixed carbonate-siliciclastic rocks (Fu, 2010; Tang et al., 2013). The exploration in the Nanyishan area has highlighted the commercial viability of low-permeability hydrocarbon reservoirs within the Paleogene Xiaganchaigou Formation. A series of exploratory wells, including Yiping 1, Yi Tan 1, Yiping 2, and Yi Tan 2, have been drilled in this region. Notably, the Yiping 1 well has demonstrated significant hydrocarbon potential, yielding daily oil production rates between 1.93 and 2.17 cubic meters, alongside daily gas production ranging from (0.85-1.46) × 10^4 cubic meters. Cumulatively, the test phase has produced 81.99 × 104 cubic meters of gas and 73.18 cubic meters of oil, highlighting the exploration potential of this area (Wan et al., 2015; Z. Zhang et al., 2023).

Previous studies have predominantly focus on the permeability classification of pore types, the interconnectivity of pores, and the processes governing the formation and evolution ofpore structures (Feng et al., 2013; Qin et al, 2018; Zhang et al., 2020). As exploration and development progress, understanding and predicting the detailed petrophysical properties of these reservoirs becomes critical for optimizing resource extraction and improving reservoir management strategies. Specifically, porosity plays an important role in this area as the lacustrine mixed carbonate-siliciclastic sequences have been found to have high porosity and low permeability, which are significant for hydrocarbon accumulation (Zhang et al., 2018). The mineralogy of these sequences influences the types of pores and their connectivity, which in turn affects the migration and trapping ofhydrocarbons (Xiong et al., 2024).

Nevertheless, these reservoirs are characterized by significant heterogeneity, which complicates their development and necessitates detailed analysis of lithology, physical properties, and pore-throat characteristics. The traditional methods for studying petrophysical properties, including core analysis and well logging, provide useful data but are often limited by high cost, difficulties in preparing samples with varying characteristics or accurately reproducing complex reservoir conditions (Pan et al., 2015; Wang et al., 2016; Zhou et al., 2016). In the meantime, machine learning have been proposed to analyze, understand and predict reservoir properties of conventional and unconventional reservoirs, enabling the use of existing data to identify complex interrelationships among key parameters (Chao et al., 2018; Schuetter et al., 2018;

Wu et al., 2014; Zhao et al., 2018). Unlike traditional empirical equations, which are often constrained by limited input variables and low forecasting accuracy, machine learning models provide a powerful means of developing predictive frameworks with higher precision and broader applicability. By integrating predictive modeling of porosity with mineralogy into reservoir management workflows, it is expected that uncertainty can be reduced in resource evaluation and optimize exploration and production strategies in the Qaidam Basin (Jun & Liang, 2024).

This study investigates the petrophysical properties of the Yiping 1 Well in the Qaidam Basin, using an existing dataset that includes thin sections, mercury intrusion, density, porosity, and permeability measurements. By applying predictive modeling techniques, the study examines the interrelationships between these properties to establish a data-driven framework for reservoir quality evaluation. The objective of this paper is to leverage the linear regression and machine learning algorithms to provide a more systematic and accurate analysis of the reservoir's petrophysical characteristics. To forecast porosity, the predictive models were developed based on mineral composition and other relevant parameters. The results could offer valuable insights into the unique features of tight carbonate reservoirs in the Qaidam Basin, which can potentially enhance the efficiency of exploration and development in the region by offering more reliable predictions and better resource management strategies.

Geological setting

The Qaidam Basin, one of China's largest inland sedimentary basins, is located on the northern Qinghai-Tibetan Plateau in northwestern China, covering an area of approximately 120,000 km^2 (Tian et al., 2022). Bounded by the East Kunlun Mountains to the south, the Qilian Mountains to the north, and the Altun Mountains to the west (Fig. 1A) (Li et al., 2020; Fu et al., 2015), it has been shaped by continuous plateau uplift and peripheral strike-slip orogeny since the Cenozoic era. These tectonic processes have resulted in an irregular diamond-shaped basin (Fig, 1B), wide in the west and narrowing toward the east (Liu et al., 2024). Renowned for its substantial petroleum and natural gas reserves, the Qaidam Basin is a key area for hydrocarbon exploration (Meng et al., 2024).

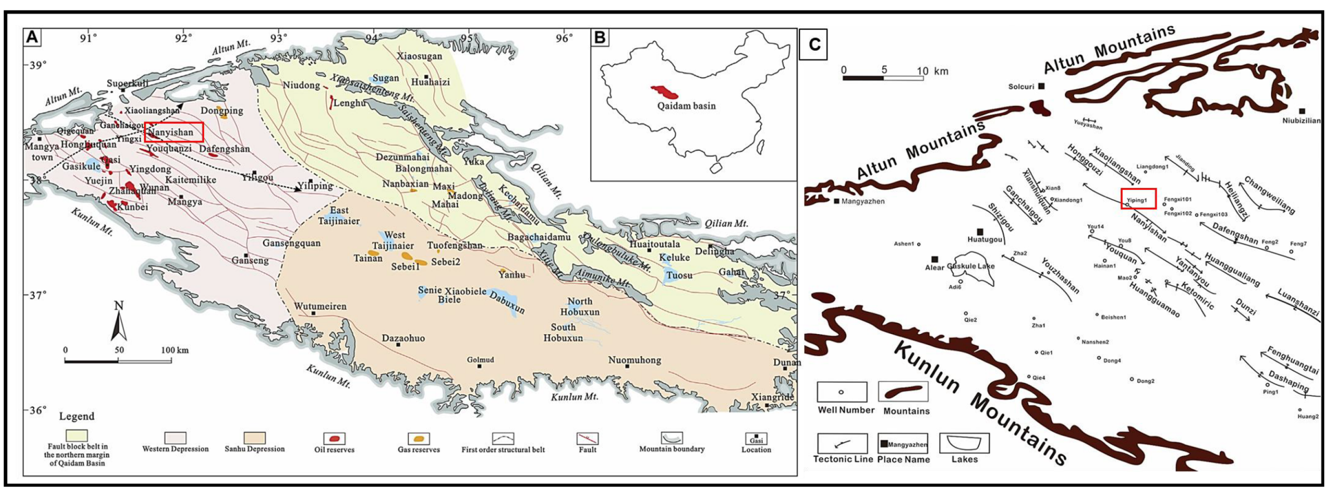


Figure 1—The location and structural map of the Qaidam Basin (Adapted from Chen et al., 2021; Liu, Xue, et al., 2024; Tang et al., 2021).(A) The pink area highlights the Western Depression, which is the main domain for oil and gas extraction, consisting predominantly of Cenozoic strata. The cyan area indicates the northern marginal fault block belt of the basin, which is the primary area for Mesozoic gas extraction. The flesh-colored area represents the Quaternary biogenic gas production zone within the basin. The two lines with arrows indicate the approximate positions of the two cross-well profiles, oriented in the NW-SE and SW-NE directions, the sketch map of the study area location, the Nanyishan is marked with a red rectangle; (B) The geographical location of the Qaidam Basin. (C) The sketch map of the study area location, the Yiping 1 Well is marked with a red rectangle.

The stratigraphic sequence in the Qaidam Basin (Fig. 2) spans the Paleogene and Neogene periods and is shaped by tectonic activity that caused the sedimentary center to migrate eastward during the Cenozoic. The Paleogene sequence consists of the Lulehe Formation at its base, followed by the Lower Ganchaigou Formation, and the Upper Ganchaigou Formation, while the Neogene sequence includes the Shangganchaigou Formation, the Lower Youshanshan Formation, the Upper Youshanshan Formation, and the Shizigou Formation (Liu et al., 2024; Wu, et al., 2022; Meng et al., 2024). The study area, as represented by Yiping 1 Well of Qinghai Oilfield, situated in the northwestern part of the Qaidam Basin (Fig. 1C). It lies on the Nanyishan anticline tectonic belt, a prominent third-order structure that serves as a key structural trap for hydrocarbon accumulation (Zhang et al., 2023). The Nanyishan area is predominantly characterized by lake-facies carbonate rocks which its reservoirs composed of argillaceous micritic dolostone, argillaceous micritic limestone, micritic limestone, and algal limestone (Gan et al., 2002). Geographically, it is surrounded by the Xiaoliangshan and Honggouzi structures to the northwest, the Dafengshan and Jiandingshan structures to the northeast, the Youquanzi structure to the southeast, and the Ganchaigou structure to the southwest (Chen et al., 2020; Zhang et al., 2023). Within this geologically complex setting, the the interplay of source rocks, reservoirs, and cap rocks in the region forms an excellent petroleum system, despite challenges related to reservoir heterogeneity and tight lithologies. In addition, the diagenetic evolution of these reservoirs is influenced by a variety of minerals, each with distinct mechanisms affecting porosity (Gan et al., 2002), resulting in porosity intricacy.

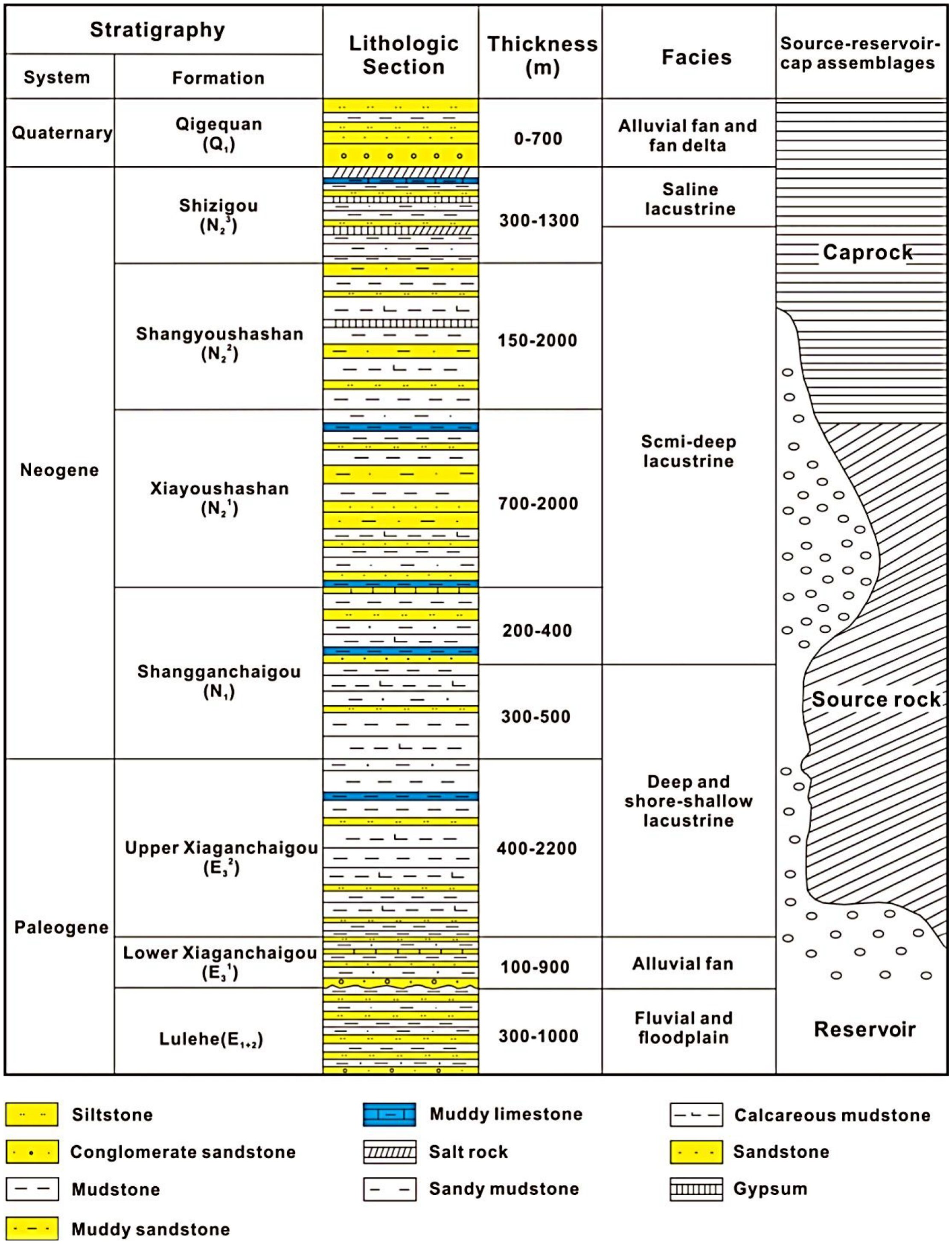


Figure 2—The Paleogene-Neogene stratigraphic column map of the western Qaidam Basin (Adapted from Chen et al., 2020).

Dataset analysis

Data source

In this study, the geological information was obtained from a continuous coring, with a total length of 65.2 meters spread across 2238 meters to 3031 meters. A total of 159 datasets were compiled for analysis: 151 were derived from X-ray diffraction (XRD) analyses across different stratigraphic formations, using an Empyrean X-ray diffractometer which is adhering to the SY/T5163-2010 standard. The stratigraphic distribution of the samples was comprised of 89 samples from the Shangganchaigou Formation, N1, 26 samples from the Xiaoyoushashan Formation, $N2^1$, and 44 samples from the Upper Xiaogancaigou Formation, $E3^2$. Additionally, 109 core samples were used to evaluate key petrophysical properties, including porosity, density, and permeability. The assessment was carried out by utilizing an Ultra Pore-400 helium porosimeter for porosity measurements and a DX-07G permeability determinator for permeability evaluations, in compliance with the GB/T29172-2012 standard. This measurement dataset was compiled of: $N2^1$ contributing 9 samples, N1 providing 64 samples, and $E3^2$ offering 36 samples (Appendix 1).

Exploratory data analysis

Relationship between density and porosity. The scatter plot presented in Fig. 3 illustrates the relationship between density mineral content (g/cm^3) and porosity (%) based on 103 effective data points collected from Yiping 1 Well.

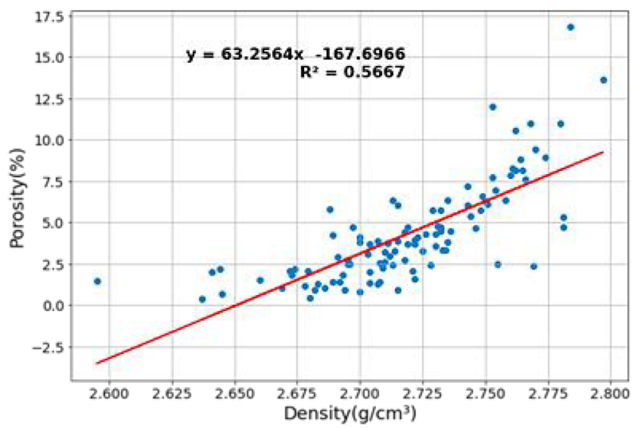


Figure 3—Relationship between content of dolomite mineral and porosity, which shows that the reservoir porosity is linearly positively correlated

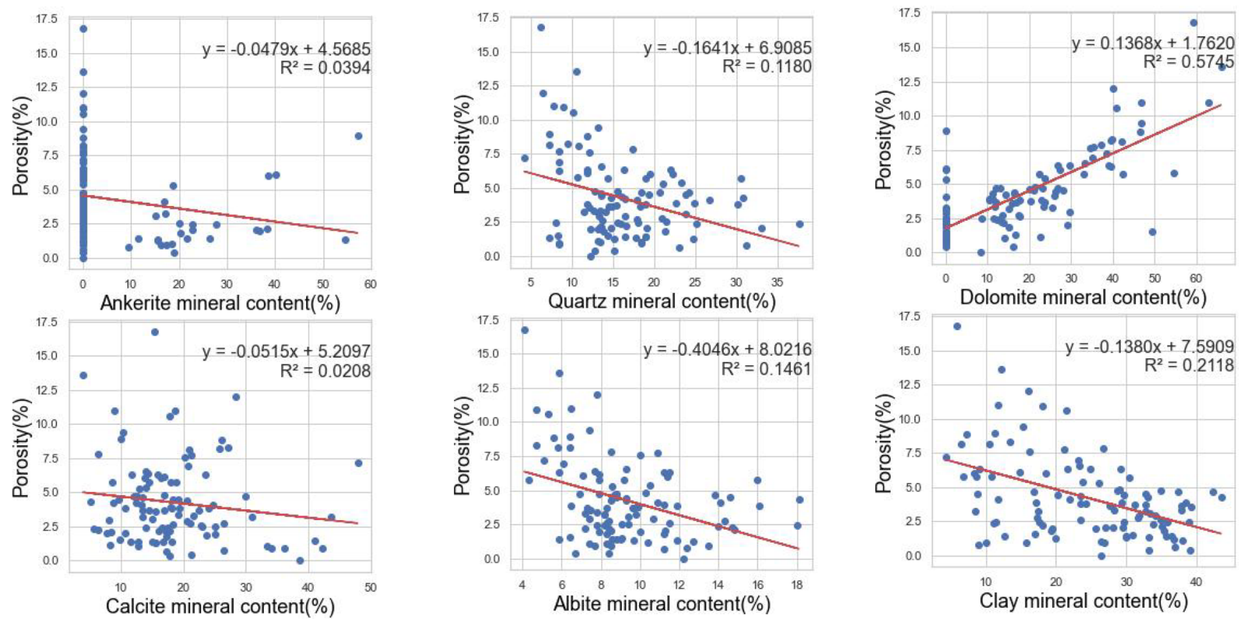
The blue dots represent individual measurements, while the red line denotes the best-fit linear regression line, characterized by the equation y=63.2564x-167.6966. The coefficient of determination R^2, calculated at 0.5667, signifies that approximately 56.67% of the variation in porosity can be attributed to changes in density mineral content (He et al., 2023). This moderate R^2 value suggests a statistically significant yet imperfect linear association, indicating that other factors might also influence porosity. The positive slope of the regression line implies that as the density mineral content increases, the porosity tends to increase correspondingly, albeit with some scatter around the trend line. Overall, this analysis offers valuable insights into the complex interactions governing porosity in reservoirs, highlighting the importance of considering multiple variables when modeling such phenomena.

Relationship between major mineral composition and porosity. Table 1 shows 6 major minerals for 151 core samples of each zone of the Yiping1 well. In all, the content of Quartz varies between 13.9% and 17.7%, with an average composition of 15.7%. Albite is present in quantities ranging from 7.9% to 10.3%, with an average of 9.1%. Calcite composition spans from 16.2% to 20.2%, averaging at 18.0%. Dolomite content is found between 17.4% and 31.4%, with an average of 25.2%. Ankerite concentrations range from 19.2% to 25.7%, with an average of 22.7%. Lastly, Clay content varies from 23.4% to 27.6%, with an average of 25.9% across the sampled cores.

Table 1—Measurements of XRD of major minerals in Yiping 1 Well

Sample Depth Range /m	Formation	Major Minerals (%)					
		Quartz	Albite	Calcite	Dolomite	Ankerite	Clay
2238.00-2247.20	$N2^1$	15.5	9.6	16.2	29.2	25.2	26.6
2655.00-2673.40	N1	15.7	8.7	18.4	23.1	20.8	27.6
2673.70-2692.30	N1	13.9	7.9	20.2	31.4	19.2	23.4
3012.00-3031.00	$E3^2$	17.7	10.3	17.2	17.4	25.7	26.0

In the regression analysis conducted on the Yiping 1 Well, the relationship between reservoir porosity and individual mineral contents was examined in Fig. 4.



(A) (B) (C)

Figure 4—Relationship between content of major minerals and porosity.(A) Relationship between the two mineral contents (ankerite, calcite) and porosity showing that the reservoir porosity is very weakly negatively correlated; (B) relationship between the two mineral contents (quartz, albite) and porosity showing that the reservoir porosity is weakly negatively correlated; (C) relationship between content of dolomite mineral and porosity showing that the reservoir porosity is linearly positively correlated, and relationship between content of clay mineral and porosity showing that the reservoir porosity is linearly negatively correlated (Wang et al., 2019).

Table 2 shows that with the exception of Dolomite, yielded R^2 values less than the critical value of $Ra^2(0.001, 104\text{-}1\text{-}1) = 0.3181$, as determined by referencing the critical value table for correlation coefficients. This indicates that the correlation analysis between the reservoir porosity and the content of individual minerals in Yiping 1 Well is not statistically reliable. The diagenetic evolution of reservoir porosity is a complex process influenced by various minerals, each contributing to porosity through different mechanisms (Xu et al., 2021). To better understand the impact of these minerals on porosity, it is essential to incorporate additional variables into the analysis and to employ more sophisticated predictive models.

Table 2—Porosity vs. mineral content linear regression analysis results

Mineral content	Fitting equation	R2	Ra²(0.001,104-1-1)
Calcite	y=-0.0515x + 5.2097	0.0208	0.3181
Dolomite	y=0.1368x + 1.7620	0.5745	0.3181
Ankerite	y=-0.0479x + 4.5685	0.0394	0.3181
Quartz	y=-0.1641x + 6.9085	0.1180	0.3181
Albite	y=-0.4046x + 8.0216	0.1461	0.3181
Clay	y=-0.1380x + 7.5909	0.2118	0.3181

Data training and testing

The dataset was randomly divided into a training set of 80% and a testing set of 20% in order to evaluate the performance of the regression models. Each data point was assigned a unique identifier (Sun et al., 2022). In this study, the density and dolomite content were selected as primary input parameters, with additional mineral composition features, including clay, quartz, calcite, and albite, considered as potential input candidates. Ultimately, density and dolomite were used as the key predictors, with porosity serving as the target variable for regression analysis.

To ensure compatibility with the requirements of the machine learning models, the dataset underwent preprocessing using both standardization and normalization techniques (Kiran et al., 2022). Standardization is mathematically defined as:

$$Z = \frac{(x - \mu)}{\sigma} \tag{1}$$

where x is the original feature value, μ is the mean of the feature values, and σ is the standard deviation. This process centers the feature at zero mean with a unit standard deviation, which is particularly beneficial for algorithms that assume Gaussian distributions, such as Support Vector Machines (SVM) and linear models. Normalization is mathematically defined as:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{2}$$

where x_{norm} is the normalized value, x is the original value, and x_{min}, x_{max} are the minimum and maximum value. This scaling brings the feature values into the range [0, 1], which is critical for algorithms that are sensitive to the absolute values of the features (Johan et al.,2018).

K-fold Cross-Validation (K-CV) is a statistical technique used in this study to rigorously evaluate the model's predictive power on new data. By iteratively training the model on K-1 subsets and validating it on the remaining subset, K-CV not only provides a reliable performance metric but also aids in hyperparameter tuning to prevent overfitting, ensuring the model's robustness and generalizability as presented in this study (S. Yadav et al., 2016).

Machine learning model development

Linear Regression (LR)

In this study, LR was employed to predict porosity based on mineral composition and density. This approach assumes a linear relationship between the predictors and the target variable, providing a baseline for model performance.

Ridge Regression (RR)

RR was utilized for porosity prediction, incorporating mineral composition and density as input features (Mcdonald G C et al., 2010). Preprocessing included standardization of the input data using the Standard Scaler, followed by feature selection through Recursive Feature Elimination (RFE) with a linear regression estimator to retain the two most relevant features. Additionally, polynomial features were generated to account for potential non-linear relationships within the data. The model was fine-tuned using five-fold cross-validation (K = 5) to identify the optimal regularization parameter (X), thereby ensuring a balance between minimizing bias and preventing overfitting.

Random Forest (RF)

RF, consisting of 100 decision trees, was trained on the original training dataset to predict porosity (Breiman et al., 2001). The model leverages ensemble learning to reduce overfitting and improve prediction accuracy. It was subsequently evaluated by generating predictions on the test dataset.

Support Vector Regression (SVR)

SVR was employed to model porosity, with preprocessing involving feature scaling to normalize the input data and ensure compatibility with the SVR algorithm (Basak D et al., 2007). A grid search, combined with five-fold cross-validation (K = 5), was conducted to optimize hyperparameters, including the penalty parameter (C) and the kernel coefficient (γ) for the radial basis function (RBF) kernel. This approach aimed to enhance the model's ability to capture complex, non-linear relationships within the data.

XGBoost (XGB)

XGB was applied to predict porosity, leveraging its gradient boosting framework to iteratively minimize prediction errors. Hyperparameter optimization was performed using a grid search over a predefined parameter space to identify the optimal model configuration. The algorithm's ability to handle complex data relationships and reduce overfitting through regularization made it a strong candidate for this task (Chen et al., 2016).

Performance evaluation

The performance of the developed linear regression and advanced machine learning algorithms was evaluated utilizing the following metrics: the coefficient of determination R2, mean squared error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) (Rainio et al., 2024).

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n} \left(y_i - \overline{y}\right)^2} \tag{3}$$

Where y_i represents the measured value, $\overline{y}$ represents the average measured value, $\hat{y}_i$ the predicted value. R^2 measures the proportion of variance in the observed data that is explained by the model, with a value closer to 1 indicating better predictive performance.

$$MSE = \frac{\left|y_i - \hat{y}_i\right|}{n} \tag{4}$$

Where n represents the number of sample data. MSE measures the average of the squared differences between the observed and predicted values, with lower values indicating better model accuracy.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}{n}} \tag{5}$$

RMSE provides an estimation of the magnitude of error in the original units of the data, with lower values indicating better predictive performance.

$$MAE = \frac{\sum_{i=1}^{n} \left| y_i - \hat{y}_i \right|}{n} \tag{6}$$

MAE measures the average of the absolute errors between the observed and predicted values, providing a simple interpretation of prediction accuracy. Lower values indicate better accuracy.

$$MAPE = \frac{\sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|}{n} \tag{7}$$

MAPE expresses the prediction error as a percentage of the actual values, making it easier to compare models across datasets with different scales. Lower values indicate better predictive performance.

These indicators provided a comprehensive quantification of the models' predictive accuracy and reliability.

Results and discussions

Predictive porosity and measured porosity

The study evaluates the efficacy of LR, RR, PR, RF, SVR, and XGB models in predicting porosity values. The R^2 was utilized to assess the extent to which each model accounted for the variability observed in the porosity measurements (Eberle et al., 2024).

Fig. 5 presents the R^2 performance of each model and RF emerged as the most accurate, with an R^2 of 0.9046, explaining approximately 90.46% of the variance in the dataset. This suggests that RF is highly effective at capturing the complex relationships between the features and the target variable, making it the best performer for porosity prediction. XGB also demonstrated strong predictive capability, with an R^2 of 0.8851, closely trailing RF (Pan et al., 2022). This model, based on gradient boosting, captured a significant portion of the variance and provided competitive performance.

In comparison, SVR performed well with an R^2 of 0.8774, while PR and RR showed moderate performance, with R^2 values of 0.8634 and 0.8211, respectively. Notably, LR demonstrated the lowest predictive power, achieving an R^2 of 0.7612, explaining only 76.12% of the variance.

The performance differences highlight that more complex models, such as RF and XGB, significantly outperform simpler approaches like LR. These findings underline the superior ability of ensemble and boosting methods to capture intricate patterns in the data, which simpler linear models may struggle to identify.

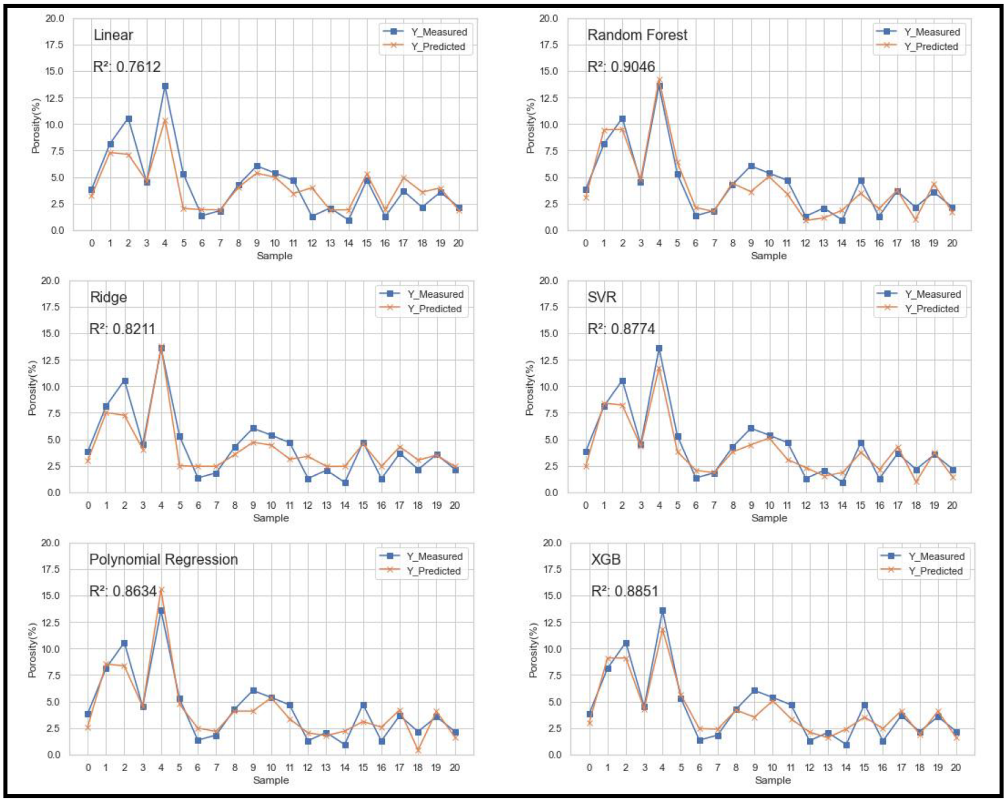


Figure 5—Measured and predictive porosity for the 6 models.

Residual error analysis

The boxplot, as illustrated in Fig. 6 provides a visual comparison of the residuals from six different machine learning models on the same dataset. Each boxplot represents the distribution of residuals for a model, with red lines indicating the median, and blue boxes encompassing the interquartile range (P10 to P90), highlighting the variability and central tendency of the prediction errors (M. Khrustalev et al., 2020).

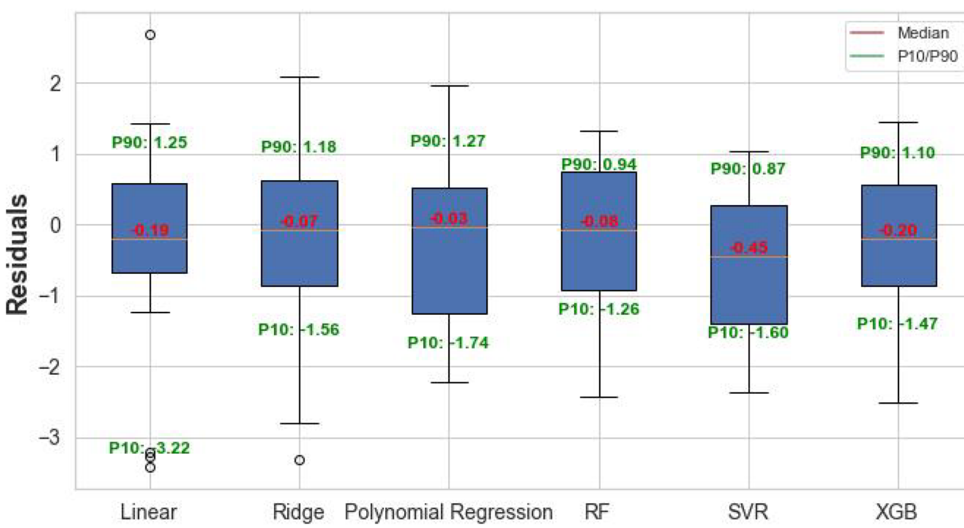


Figure 6—Comparative residual analysis of six machine learning models

A deeper analysis of the residual errors further underscores the robustness of RF and SVR. RF exhibited the most consistent performance, with a median residual of −0.08, a narrow interquartile range (IQR), and tight bounds (P10: −1.26, P90: 0.94). These metrics suggest that RF not only produces accurate predictions but also maintains consistency across the dataset.

SVR, while slightly more biased with a median residual of −0.45, demonstrated good consistency as well, with the smallest P90 value of 0.87, though it exhibited a slight negative bias in its predictions. Polynomial Regression achieved the smallest median residual (-0.03) and demonstrated robust control of the lower bound (P10: −1.74), although variability increased at the upper bound (P90: 1.27), indicating less consistency compared to RF. Similarly, Ridge Regression performed comparably to PR but exhibited a slightly wider spread in residuals (P10: −1.56, P90: 1.18).

On the other hand, LR exhibited the poorest residual performance, with the largest residual range (P10: −3.22, P90: 1.25), along with significant outliers, indicating high inconsistency in predictions. This suggests that the simple linear approach may not be well-suited for the non-linear relationships in the data.

XGB showed moderate performance with a median residual of −0.20, and P10 and P90 values of −1.47 and 1.10, respectively. While it exhibited more variability than RF and SVR, it still provided competitive results in terms of residual consistency. Overall, RF and SVR outperform simpler models in predictive accuracy and consistency, while LR struggles with variability and outlier sensitivity.

Comparison of prediction accuracy among 6 models

Table 3 presents the performance evaluation metrics of six regression models, with performance metrics. In terms of overall prediction accuracy, the RF model outperformed all other algorithms, with the lowest MSE of 0.93, RMSE of 0.97, and MAE of 0.81. These metrics suggest that RF provides the most reliable predictions with minimal errors. Additionally, RF achieved the highest R^2 value of 0.9046, confirming its strong ability to explain the variance in the data. Its MAPE was also the lowest (27.41%), indicating that the relative prediction error is minimal compared to the other models.

Table 3—Performance evaluation metrics of six models

Model	MSE	RMSE	MAE	MAPE (%)	R^2
LR	2.34	1.53	1.10	35.73	0.76
RR	1.75	1.32	1.03	39.94	0.82
PR	1.34	1.16	0.94	34.65	0.86
RF	0.93	0.97	0.81	27.41	0.90
SVR	1.20	1.10	0.91	30.57	0.88
XGB	1.12	1.06	0.88	32.67	0.89

In contrast, LR showed the poorest performance with the highest MSE (2.34), RMSE (1.53), and MAE (1.10). This resulted in the lowest R^2 value of 0.7612, suggesting that the LR model is least effective at capturing the underlying relationships in the data. The relatively high error metrics indicate that LR struggles with prediction accuracy, especially for more complex datasets like the one in this study.

RR and PR showed better performance than LR but still fell short of RF. RR demonstrated moderate error metrics and an R^2 of 0.8211, while PR showed a slightly higher R^2 of 0.8634, indicating its ability to handle non-linearities in the data better than Ridge Regression. SVR and XGB performed well with moderate error metrics and high R^2 values. The results for these models suggest that they are competitive alternatives to RF, providing strong predictions while balancing bias and variance.

For application purpose, RF has been applied to predict the porosity values and Table 4 demonstrates the measured porosity and predicted porosity. With density and dolomite as the fixed variables, the values of predicted porosity are close to measured porosity. For example, Sample 5 has a high measured porosity of 13.597%, which is closely predicted by the model at 14.209%, indicating a high degree of accuracy. However, there are noticeable differences between measured and predicted values in some samples. For instance, Sample 10 has a measured porosity of 6.038%, while the model predicts a lower value of 3.617%. Such discrepancies might be due to the complexity ofthe rock's pore structure which requires further investigation.

Table 4—Measured porosity and predicted porosity of random forest model

Sample Number	Density	Dolomite (%)	Measured Porosity (%)	Predicted Porosity (%)
1	2.70	10.40	3.819	3.070
2	2.76	42.00	8.125	9.458
3	2.76	40.70	10.569	9.481
4	2.73	21.00	4.506	4.779
5	2.80	65.90	13.597	14.209
6	2.78	0.00	5.298	6.394
7	2.704	0	1.348	2.112
8	2.693	0	1.826	1.747
9	2.73	16.4	4.256	4.424
10	2.715	27.2	6.038	3.617
11	2.744	23.8	5.378	5.041
12	2.719	11.9	4.679	3.411
13	2.683	16.6	1.296	0.894
14	2.679	0	2.074	1.141
15	2.694	0	0.936	1.875
16	2.697	26.9	4.727	3.462
17	2.707	0	1.273	2.010
18	2.722	23.6	3.676	3.744
19	2.644	13.9	2.151	0.986
20	2.73	15.9	3.568	4.346
21	2.674	0	2.167	1.717

Conclusions and future works

This study investigates the petrophysical properties of the Nanyinshan reservoir in the Qaidam Basin, employing advanced machine learning techniques to predict porosity with high accuracy. The analysis of 151 core samples revealed significant stratigraphic differences in mineral composition and porosity, providing valuable insights into the geological characteristics of the reservoir rocks. Dolomite exhibited a strong positive correlation with porosity ($R^2 = 0.5745$), while clay showed a negative correlation ($R^2 = 0.2118$). Other minerals, including quartz, albite, calcite, and ankerite, demonstrated weak or negligible effects on porosity.

Among the six machine learning algorithms evaluated, RF emerged as the most effective model, achieving the highest R^2 value (0.9046) and the lowest error metrics (MSE: 0.93, RMSE: 0.97, MAE: 0.81). The RF model demonstrated a superior ability to capture complex patterns within the dataset and provided consistent, reliable predictions. Residual error analysis highlighted RF's robustness, with a narrow residual range and a low median residual (-0.08), outperforming other models such as XGB and SVR. LR, in contrast, exhibited the weakest performance, characterized by significant variability and sensitivity to outliers ($R^2 = 0.7612$).

A moderate linear relationship between density and porosity ($R^2 = 0.5667$) was identified, indicating that density partially influences porosity. However, the moderate R^2 suggests that additional factors, beyond density, contribute to porosity variations. The findings underscore the importance of advanced machine learning models in capturing these complex relationships and enhancing the accuracy of reservoir characterization.

This research highlights the potential of integrating machine learning techniques, particularly ensemble and boosting methods, into traditional reservoir analysis workflows. Such integration can improve the precision of reservoir evaluations, supporting more efficient exploration and production strategies. Future work should focus on expanding the input dataset to incorporate well logs, seismic attributes, and other petrophysical measurements to enhance the generalizability of the models. Additionally, exploring hybrid modeling approaches and optimizing hyperparameters could further improve predictive performance. These advancements will contribute to a deeper understanding of tight carbonate reservoirs and support data-driven decision-making in exploration and development activities within the Qaidam Basin and similar analogous geological settings.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

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Appendix

Appendix 1—Measurements for 159 core samples of the well.

Sample ID	Formation	Sample Depth/m	Petrophysics	XRD
1	N_2^1	2238.7	√	√
2	N_2^1	2238.97	√	√
3	N_2^1	2239.28	√	√
4	N_2^1	2239.38	√	√
5	N_2^1	2241.38	√	√
6	N_2^1	2242.2		√
7	N_2^1	2242.4		√
8	N_2^1	2243	√	√
9	N_2^1	2245.3		√
10	N_2^1	2246.28	√	√
11	N_2^1	2246.56		√
12	N_2^1	2246.72		√
13	N_2^1	2247.18	√	√
14	N_2^1	2240.4		√
15	N_2^1	2240.6		√
16	N_2^1	2240.65		√
17	N_2^1	2241.2		√
18	N_2^1	2241.62		√
19	N_2^1	2244.36		√
20	N_2^1	2245		√
21	N_2^1	2245.13		√
22	N_2^1	2245.4		√
23	N_2^1	2245.68		√
24	N_2^1	2246.12		√
25	N_2^1	2238.15		√
26	N_2^1	2244.15	√	√

Sample ID	Formation	Sample Depth/m	Petrophysics	XRD
27	N_1	2655.5		√
28	N_1	2655.68		√
29	N_1	2655.88	√	√
30	N_1	2656.18	√	√
31	N_1	2656.48		√
32	N_1	2656.68		√
33	N_1	2656.95	√	√
34	N_1	2657.81	√	√
35	N_1	2658.11	√	√
36	N_1	2658.4	√	√
37	N_1	2658.65	√	√
38	N_1	2659.33		√
39	N_1	2659.73	√	√
40	N_1	2659.96	√	√
41	N_1	2660.23	√	√
42	N_1	2660.47	√	√
43	N_1	2660.9		√
44	N_1	2661.36	√	√
45	N_1	2661.78	√	√
46	N_1	2661.93	√	√
47	N_1	2662.2		√
48	N_1	2663.2	√	√
49	N_1	2663.46	√	√
50	N_1	2664.08	√	√
51	N_1	2664.29	√	√
52	N_1	2664.78	√	√
53	N_1	2665.42		√
54	N_1	2665.81	√	√
55	N_1	2667.19	√	√
56	N_1	2667.39	√	√
57	N_1	2667.55	√	√

Sample ID	Formation	Sample Depth/m	Petrophysics	XRD
58	N_1	2667.72	√	√
59	N_1	2667.94	√	√
60	N_1	2668.4	√	√
61	N_1	2668.78	√	√
62	N_1	2669.73	√	√
63	N_1	2670.12	√	√
64	N_1	2670.28	√	√
65	N_1	2670.63	√	√
66	N_1	2671.07	√	√
67	N_1	2671.17	√	√
68	N_1	2671.44	√	√
69	N_1	2671.52	√	√
70	N_1	2671.93	√	√
71	N_1	2672.54	√	√
72	N_1	2672.62	√	√
73	N_1	2672.79	√	√
74	N_1	2672.98	√	√
75	N_1	2674.14	√	√
76	N_1	2674.24	√	√
77	N_1	2674.7		√
78	N_1	2675.8	√	
79	N_1	2676.04	√	√
80	N_1	2676.17	√	√
81	N_1	2676.4	√	√
82	N_1	2679.44	√	√
83	N_1	2680.16	√	√
84	N_1	2680.38	√	√
85	N_1	2680.7	√	√
86	N_1	2681.04		√
87	N_1	2684.03	√	√
88	N_1	2684.12	√	√

Sample ID	Formation	Sample Depth/m	Petrophysics	XRD
89	N_1	2684.32	√	√
90	N_1	2684.55		√
91	N_1	2685.1		√
92	N_1	2686.53	√	√
93	N_1	2686.7	√	√
94	N_1	2686.83		√
95	N_1	2687.45	√	√
96	N_1	2687.65	√	√
97	N_1	2688.03		√
98	N_1	2688.66		
99	N_1	2688.9		√
100	N_1	2689.09	√	√
101	N_1	2689.26	√	√
102	N_1	2690.33	√	√
103	N_1	2657.22		√
104	N_1	2661.56		√
105	N_1	2664.44		√
106	N_1	2668.43		√
107	N_1	2668.62		√
108	N_1	2669.12		√
109	N_1	2669.97		√
110	N_1	2662.5		√
111	N_1	2658.75	√	
112	N_1	2662.8		
113	N_1	2671.2	√	√
114	N_1	2672.15	√	√
115	N_1	2688.7	√	√
116	E_32	3012.28	√	√
117	E_32	3012.59	√	√
118	E_32	3012.88	√	√
119	E_32	3013.32	√	

Sample ID	Formation	Sample Depth/m	Petrophysics	XRD
120	E_32	3013.6		√
121	E_32	3013.96	√	√
122	E_32	3014.57	√	√
123	E_32	3014.81	√	√
124	E_32	3014.98		
125	E_32	3015.15	√	√
126	E_32	3015.37	√	√
127	E_32	3015.9	√	√
128	E_32	3018.46	√	√
129	E_32	3019.5	√	√
130	E_32	3019.64	√	√
131	E_32	3019.88	√	√
132	E_32	3020.05	√	√
133	E_32	3020.3	√	√
134	E_32	3020.48	√	√
135	E_32	3020.6	√	√
136	E_32	3020.88		√
137	E_32	3021.24	√	√
138	E_32	3021.34	√	√
139	E_32	3021.56		√
140	E_32	3021.63	√	√
141	E_32	3021.98	√	√
142	E_32	3022.2	√	√
143	E_32	3022.36	√	√
144	E_32	3022.85	√	
145	E_32	3023.14	√	√
146	E_32	3023.79	√	√
147	E_32	3024.03		√
148	E_32	3024.2	√	√
149	E_32	3026.3	√	√
150	E_32	3026.89	√	√
151	E_32	3026.93		√

Sample ID	Formation	Sample Depth/m	Petrophysics	XRD
152	E_32	3028.17	√	√
153	E_32	3028.31	√	√
154	E_32	3029.22	√	√
155	E_32	3029.43	√	√
156	E_32	3029.7		√
157	E_32	3030		√
158	E_32	3030.38	√	√
159	E_32	3030.8	√	